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Silica and Silicate Scales Formation in Geothermal Condition and Their Control

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Abstract

Silica and metal silicate formation in the presence of divalent metal ions such as magnesium, calcium, iron, and aluminum, along with dissolved silica, is commonly observed in geothermal systems at 150 to 350°C. Even though geothermal systems for power plants can reach temperatures in the range of 200-300°C or even higher, most silica solubility and inhibition studies reported in the literature were performed at low enthalpy geothermal systems up to 100°C in laboratory conditions due to testing challenges at high temperatures. To address the gap, we employed a range of analytical methods to evaluate silica and metal silicate formation under ultra-high temperature conditions (originally up to 200°C and more recently up to 250°C). We also assessed the performance of scale inhibitors suitable for these extreme conditions.

This study introduces two experimental approaches: a static bottle test and a high-temperature high-pressure (HTHP) flow-through system, which were developed to study silica and metal silicate formation and inhibition. The static bottle testing method, suitable for low to medium enthalpy geothermal systems, operates up to 150°C. The HPHT flow-through work method for silica and silicate using DSL was designed for temperatures up to 200°C, and a modified HTHP flow-through version was proposed for testing at 250°C.

In this study, we present novel chemical solutions to control silica and silicate scales and qualified several chemistries using both testing methods. The inhibitors were tested against the complex brine prone to form both calcite and silicate scales. The testing results indicate that several formulations effectively inhibit and disperse the silica & silicate scales at ultra-high temperatures. Findings highlight the sensitivity of soluble silica concentration to slight variations in temperature, pH, and the presence of inhibitors or chelators, all of which significantly influence silica solubility and/or precipitation kinetics

INTRODUCTION

Silica and metal silicates are common inorganic scale deposits found in various aqueous systems, including oil and gas production, industrial water treatment, and geothermal operations. These deposits often co-occur with other mineral scales such as calcium carbonate, calcium sulfate species, barium sulfate, and calcium phosphates (Zhul and Amjad 2014). These scale deposits typically occur when brines become supersaturated with ions such as Ba^{2+}, Ca^{2+}, Sr^{2+}, SiO_4^{-} /SiO_3^{2-}, and SO_4^{2-}. This supersaturation can result from changes in thermodynamic conditions, such as temperature, pressure, and pH, or from operational factors like the mixing of waters from different zones, acidification treatments, Alkaline-Surfactant-Polymer (ASP) flooding, or Water-Alternating-Gas enhanced well recovery (WAG EOR) (Paudyal, et al. 2022). These excess ions precipitate from the system as a function of solubility equilibrium (Kan and Tomson 2012).

Silica and metal silicate scaling is also a concern in industries that rely on sourced or feed water, including evaporative cooling systems, semiconductor manufacture, and boilers (Demadis, Skordalou and Aristodemou 2020). Silica and metal silicates typically form in amorphous form and pose significant challenges due to their complexity, which varies with brine composition. Geographically, such scaling issues are prevalent in regions like Texas, Arizona, New Mexico, and some parts of California in the US, southern parts of Europe, the Pacific Rim, and Latin America (Demadis, Skordalou and Aristodemou 2020) (Henley 1983).

Among the various metal silicates, magnesium silicate is the most encountered in oil and gas production and water purification systems (Ikuko 2022). In brines saturated with divalent metal ions such as magnesium, magnesium hydroxide can form and subsequently react with colloidal silicate particles to produce magnesium silicate. Although metal silicates can precipitate in both crystalline and amorphous forms, the amorphous phase is more prevalent due to its spontaneous formation, rapid agglomeration, and growth of the metal silicate chains (Iler 1979). This process is further accelerated at elevated temperatures, particularly above 60°C (Gunnarsson, Arnorsson and Jacobsson 2005). Once formed, magnesium silicate scales are difficult to remove due to their low solubility under these HTHP conditions. Their complexity increases in the presence of other scale-forming species such as calcium carbonate ($CaCO_3$), calcium phosphate, and metal ions like Fe^{2+}/Fe^{3+} or Al^{3+}, which contribute additional hydroxides and cations (K. D. Demadis 2003). The scale matrix may also include other divalent and monovalent cations such as Mg^{2+}, Ca^{2+}, Na^{+}, and K^{+} (Todd and Bluemle 2022). The solubility of silica and metal silicates is strongly influenced by the system pH. In geothermal and other aqueous systems, silica and metal silicate precipitation typically occurs within a pH range of 8 to 10. Within this range, silicic acid remains in protonated form, favoring the formation of monomeric silica to silicate ions [$Si(OH)_3O^-$], which readily interact with metal ions to form insoluble metal silicates (Demadis, Spinthaki, et al. 2019).

Evidence of the formation of amorphous to poorly crystalline metal silicates was reported in geothermal installations, notably in Iceland (Arnorsson 2000). Although magnesium concentrations in the groundwater from springs and drillholes used in geothermal energy generation are typically low, the mixing of hot geothermal brine of pH approximately 9 or above with such groundwater creates favorable conditions for metal silicate precipitation (Gunnarsson and Arnorsson 2000). In some other fields, such as Japan's Fushime geothermal area within the Yamagawa field on the Satsuma peninsula, SiO_2 concentration measured at atmospheric temperature is reported in the range of 540 to 1,250 mg/L, and the temperature is recorded as higher than 300°C. In this field, magnesium silicate formation has been observed in some of the wells, even at neutral pH with a moderate amount of magnesium when SiO_2 is extremely high (Yagi, et al. 2000). The amorphous scale deposits have also been found on surface equipment when the operating temperature is around 200°C. These scales typically consist of a mixture of silica, iron oxide, and alkaline earth oxide (Gallup 1989). In high-enthalpy geothermal systems, where the temperature of the reservoir brine remains in the range of 200 to 350°C, the concentration of dissolved silica in the aquifer waters is generally high (300−700 ppm). As the brine cools to 100-200°C, it becomes supersaturated due to a shift in solubility equilibrium, promoting the precipitation of amorphous silica or metal silicates (Saadi and Haddabi 2019). Such deposits are frequently observed in geothermal power plants and installations, particularly when the brine contains elevated levels of metal ions such as magnesium, calcium, iron, and aluminum (Demadis, Spinthaki, et al. 2019). The solubility of silica is highly sensitive to changes in temperature and pH, and even slight variations in these parameters can significantly impact precipitation behavior. Despite the high temperatures typical of geothermal systems, most silica solubility studies reported in the literature were conducted at temperatures ≤ 100°C. Due to the scarcity of high-temperature data, solubility predictions for silica and metal silicates at elevated temperatures are often derived using multiple linear regression models based on low-temperature data.

One of the common strategies for preventing colloidal and amorphous silica deposition is the use of chemical inhibitors and additives. Nonpolymeric inhibitors such as phosphonates and phosphate esters, as well as polymeric inhibitors including polyphosphonate, polycarboxylates, homo- or copolymers of acrylic or maleic acid, are widely used to control inorganic scales in aqueous conditions (Kelland 2014). These anionic inhibitors are particularly effective against carbonate and sulfate scales. Anionic additives containing functional groups like carboxylates, phosphonates, or both have also been applied to mitigate silica and silicates (Demadis, Mavredaki and Somara 2011). However, controlling silica-based scales remains a significant challenge. This is primarily because silica and silicates tend to form polymerized amorphous solids, which are less responsive to conventional scale inhibitors chemistry. Moreover, metal silicate formation is favored at higher pH levels (typically between 8 and 10), a range in which many standard inhibitors show reduced effectiveness. As a result, most inhibition studies conducted within this pH window have demonstrated limited success in controlling silica and silicate scale formation. Consequently, the prevention of silica and silicate precipitation and deposition on equipment surfaces in industrial water systems, especially under high-temperature and high-pH conditions, remains a complex and unresolved issue. This paper presents the development and evaluation of novel chemical formulations designed to inhibit silica, metal silicate, and calcite-silicate scale formation at ultra-high temperatures up to 200°C.

MATERIALS AND METHODS

Scale modeling with SSP

ScaleSoftPitzer (SSP) is a mathematical tool that is based on thermodynamic parameters to calculate scaling potential at various stages in the production process (Kan and Tomson 2012), (W.Shi 2012). The scale prediction tool gives an estimation of severity for types of scale, based on provided conditions and ion composition, as precipitation risk or scaling risk with the change in thermodynamic parameters such as pressure and/or temperature. Scale formation tendency is reported in terms of Saturation ratio (SR) or Saturation Index (SI). Where the $SI < 0$ or $SR < 1$, there is no risk for the formation of scale as the brine is undersaturated, at $SI = 0$ or $SR = 1$, the brine is at equilibrium and where $SI > 0$ or $SR > 1$, it is supersaturated and has a tendency to form inorganic form of precipitation. The higher the saturation index or saturation ratio, there is higher potential to form scale and is more thermodynamically favoured and therefore it is harder to control.

HTHP flow loop test method

Flow loop tests were conducted using the Dynamic Scale Loop (DSL) to evaluate the inhibition performance of selected scale inhibitors. Separate cationic and anionic brines were prepared using the ions composition as in Table 1, and the pH of each brine was adjusted to 8.5 to reflect field conditions more accurately. These brines were then pumped independently into the test rig, where they mixed at the inlet of a narrow-coiled tube (1mmX3m), simulating the conditions under which scale formation typically occurs. The tests were performed at temperatures ranging from 175 or 200°C, a constant pressure of 500 psi, and at flow rates of 2ml/min at 175°C and 10 ml/min at 200°C. Scale deposition within the coil was continuously monitored by measuring the differential pressure (#P) change across the coil. Tests were run both in the presence and absence of inhibitors. A #P increase of more than 1 psi from the baseline #P was used as the threshold limit to determine inhibitor performance − values above this threshold indicated significant scale formation and reduced inhibitor effectiveness. The developed geothermal metal silicate (GTMS) inhibitor formulations

(Table 2) were tested in two different brines, one was supersaturated with silica and magnesium silicate scales, and the other was supersaturated with silica, magnesium silicate, and calcite together (Table 1).

Table 1—Composition of metal silicate and calcite brines for tube blocking tests

ION	Metal silicate brine [mg/L]	Calcite and metal silicate brine [mg/L]
Na^+	1592	1592
Ca^{2+}	43.5	43.5
Mg	50-100	50
K^+	47	47
Sr^{2+}	1.6	1.6
Cl^-	2346	2346
SO_4^{2-}	50	50
HCO_3^-	0	180
Si	230-700	230
pH	**8.5**	**8.5**
Temp (°C)	**175-195**	**195**
Pressure (psi)	**500**	**500**

Table 2—List of products used in silica and metal silicate inhibition evaluation

Formulated Products	Chemistries
GTMS-1	Polymer
GTMS-2	Polymer
GTMS-3	Phosphonated polymer
GTMS-4	Polymer
GTMS-5	Polymer
GTMS-6	Polymer
GTMS-7	Organic acid
GTMS-8	Polymer
GTMS-9	Phosphonate
GTMS-10	Polymer
GTMS-11	Polymer

HTHP static test

The scale precipitation, deposition, and flocculation process were evaluated using a static method in 60 ml HTHP ACE glass cells (TP rating, 150 psi at 120°C) at a temperature of 150°C. Separated anionic solution and cationic solution were prepared with two times the concentration of the ions composition, with each adjusted to a pH 8.5 to reflect field-relevant conditions. The concentrations of ions such as Ca, Si, Mg, and HCO_3 were maintained as specified in Table 1. Equal volumes of the anionic and cationic solutions were mixed in a 1:1 ratio and placed in an oven at 150°C to initiate the reaction (Figure 1). For inhibitor evaluation, the anionic solution was pre-dosed with inhibitors at 2X the targeted concentration before mixing. Visual Observations were recorded at 4 hours and 24 hours to monitor scale formation. Sampling for elemental analyses using ICP-OES was performed after 24 hours when samples were relatively cooler to a manageable temperature (~75°C), and they were filtered through 0.2µm membrane filters for elemental analysis. The filtrates were diluted (~10X) and acidified with 3% nitric acid to match the matrix of standard ICP-OES procedures and prevent further precipitation. ICP-OES analysis was performed to quantify the residual composition of the targeted elements such as Ca, Mg, and Si in the filtrates. The inhibition efficiency, as the inhibition percentage, was calculated by comparing these concentrations against those in the blank and control samples, providing a quantitative measure of inhibitor performance.

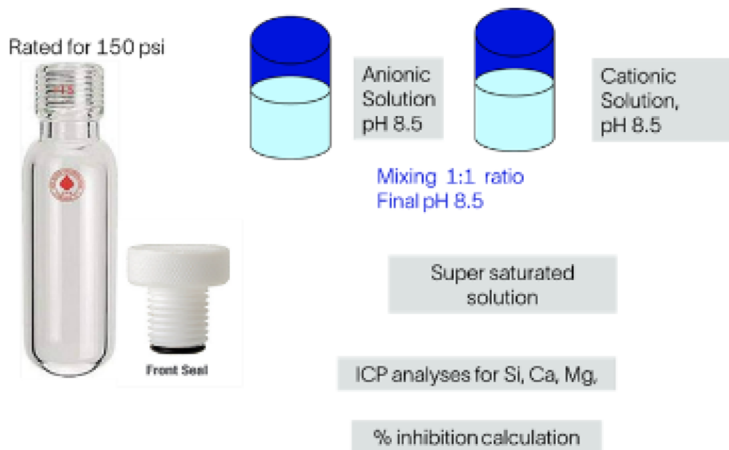

Figure 1—HTHP static test apparatus and quick procedure

RESULTS AND DISCUSSION

Silica and silicate scale risk in geothermal conditions

To assess the scaling severity of the testing brines used in this study, SSP software was used to simulate the saturation level of the silicate/ Mg-silicate and silicate/Mg-silicate/calcite brines with this drop in temperature and to predict the amount of precipitation at different temperature points from low to high enthalpy geothermal temperature. Two distinct brine compositions were considered for the modeling: one supersaturated with silica, Mg-silicate, and Ca/Mg-silicate; the other supersaturated with silica, Mg-silicate-calcite, Ca/Mg-silicate-calcite (Table 1). Simulations were conducted across a temperature range of 60-250°C, representing conditions from the surface facility to the high-enthalpy geothermal reservoirs. The pH was maintained at a constant level of 8.5, approximating the brine condition typically observed in geothermal power plants.

For the metal silicate brine, increasing the temperature from 60°C to 250°C, resulted in a rise in the predicted Saturation Indices (SI) for Mg silicate from 2.27 to 8.45 (corresponding precipitated mass: ~2.20 to 4.22 mg/L) and for Ca/Mg silicate 1.19 to 5.08 (precipitated mass ~2.21 to 4.69 mg/L) as shown in Figure 2a. When alkalinity was considered in the brine to estimate calcite supersaturation (Table 2), the predicted supersaturation level as SI for Mg-silicate was drastically increased. Even though saturation index calculations for different scales are independent, the results suggest a favorable environment for co-precipitation of metal silicates and calcite. Specifically, the SI values ranged from 7.43 to 16.48 for Mg-silicate,, 4.62 to 10.39 for Ca/Mg-Silicate,, 0.57 to 2.20for Calcite. Corresponding precipitation amounts were estimated as follows: Mg-silicate, 60.37 to 135.91mg/L; Ca/Mg-Silicate, 66 to 159.16 mg/L; and calcite, 37 to 101 mg/L(Figure 2b). It is important to note that these values are derived from empirical calculation based on thermodynamic parameters used in SSP.

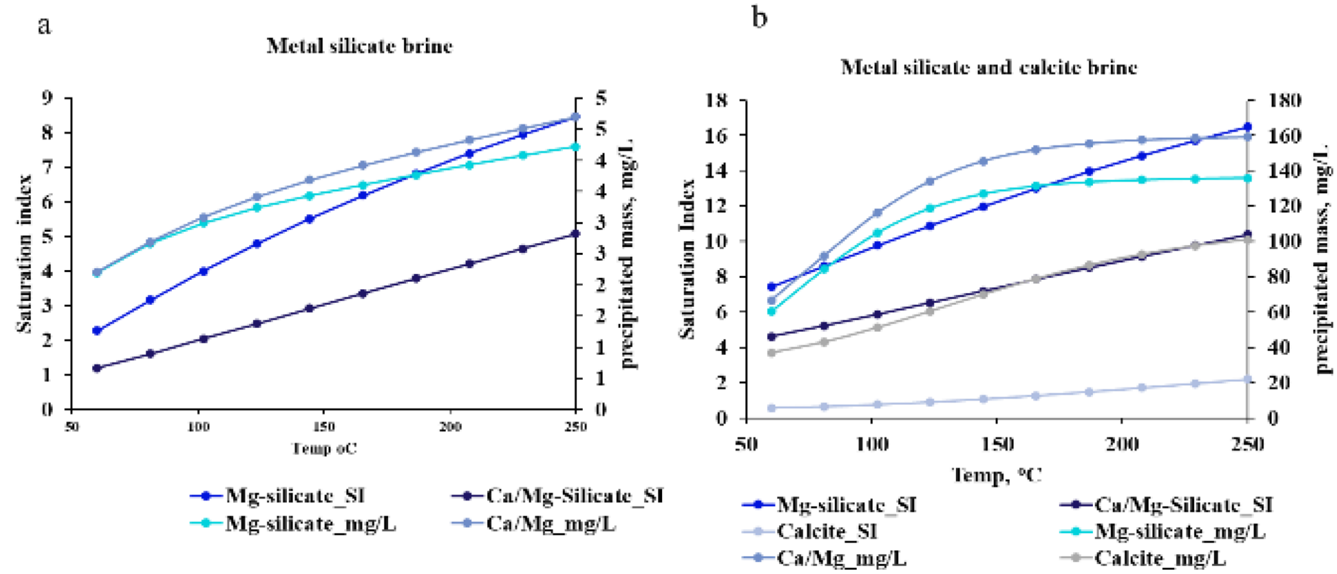

Figure 2—Silica and metal silicate scaling risk. a, Prediction of saturation indices and precipitation mass for metal silicate brine from temp 60 to 250°C b, Prediction of saturation indices and precipitation mass for metal silicate and calcite brine from temp 60 to 250°C

Inhibition of silica and metal silicates

Several potential raw materials suitable for HTHP or geothermal work were initially screened at a lower temperature of 95°C. From this screening, 11 potential candidates that include polymers, phosphonates, and phosphonated polymers were selected for further evaluation at elevated temperature (Table 2). Bottle tests were performed using the brine supersaturated with silica, Mg-silicate and Ca/Mg-silicate at 150°C, with a reaction time of 24 hours. Among the 11 candidates, only inhibitors GTMS-1, 2, 3, 4, and 11 are shown here in Figure 3. After 24 hours at 150°C, the Blank bottle (without any inhibitor) exhibited loosely formed amorphous, cloudy clusters at the bottom of the bottle. In contrast, bottles treated with 500 ppm of GTMS-1, 2, 3, 4, and 11 showed no visible precipitation during the same period. However, bottles treated with GTMS-1, 2, 4, and 11 developed amber coloration over time, which may be attributed to exposure of brine with chemicals at elevated temperatures (Figure 3, right).

Figure 3—Evaluation of inhibition of silica and magnesium silicate at 150°C with static method for 24 hours. Left, at initial time point right after mixing anion and cation in equal ration. Right, after 24 hours of reaction at 150°C

Additional tests were performed at 150°C using brine containing 50 ppm of Mg and 230 ppm of Si to understand the recovery rate of scaling ions. GTMS-1 through GTMS- 9 were individually assessed for their inhibition efficiency at a concentration of 500 ppm in the test brine (Figure 4a). Among these tested products via static method, some functioned primarily as dispersants, maintaining amorphous particles in suspension until the time of sampling, while others worked as inhibitors, resulting in visibly clear treated samples.

To quantitatively assess the inhibition efficiency, elemental analyses using ICP-OES were performed after 24 hours of reaction. Prior to analysis, the samples were filtered using 0.2µm membrane filter. It is to be noted that the sampling process was performed at a temperature below 100°C to mitigate risks associated with high temperature handling. Magnesium recovery rates and percentage inhibition were calculated based on the Mg^{2+} concentrations in treated samples, control samples, and blank samples as described in Eq.1. The ICP-OES results revealed 100% recovery of Mg^{2+} ions in the samples treated with GTMS 7 and 8, confirming their strong inhibitory properties to control metal silicates at 150°C. GTMS 7,an organic acid, is particularly effective in chelating divalent metal ions such as calcium and magnesium. Polymeric inhibitors, GTMS-1, GTMS-2, and GTMS-4 also demonstrated high efficiency in controlling silica and metal silicates at these tested conditions, with recovery rates more than 80%. In contrast, other selected inhibitors including polymers, phosphonates, and poly-phosphonates(GTMS-3, GTMS-5, GTMS-6 and GTMS-9) showed lower recovery rates despite their ability to disperse the particles. The reduced recovery rates of these products may be attributed to the formation of larger aggregated particles exceeding the pore size of the 0.2µm membrane filter, resulting in their retention during filtration.

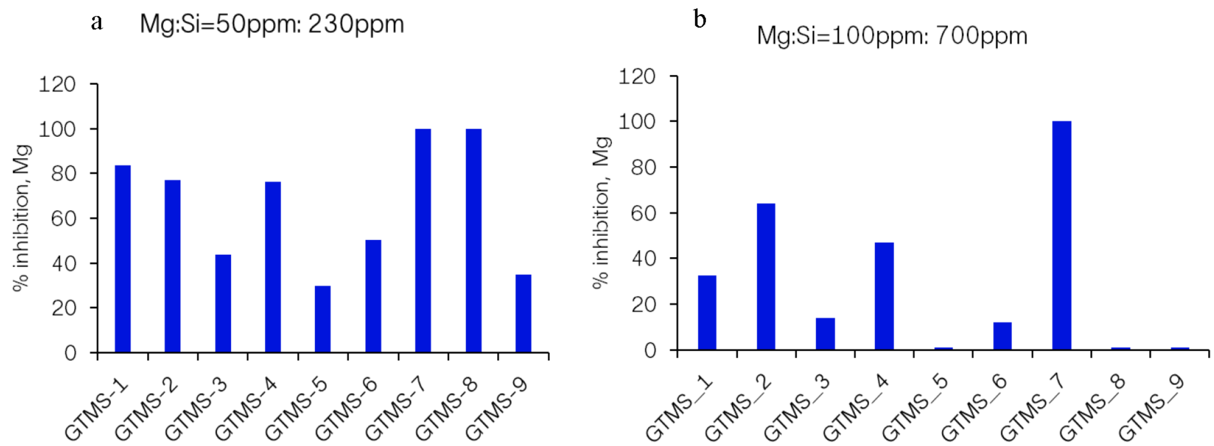

Figure 4—Evaluation of inhibition of silica and magnesium silicate at 150°C with regard to Mg concentration in suspension phase. a, Mg and Si ratio concentration is maintained at 50 ppm and 230 ppm respectively as in the original brine at Table 1. b, Mg and Si ratio concentration

$$\%\ inhibition = \frac{(Mg^{2+}Cs - Mg^{2+}Cb) X 100}{(Mg^{2+}Co - Mg^{2+}Cb)} \qquad \text{Eq. 1}$$

$Mg^{2+}{}_{Cs}$ = Concentration of magnesium ion in inhibitor treated sample at the time of sampling

$Mg^{2+}{}_{Cb}$ = Concentration magnesium ion in blank.

$Mg^{2+}{}_{Co}$ = Concentration of magnesium ion in original or control samples

Similar tests were conducted at 150°C under more severe conditions using the brine containing 100 ppm of Mg and 700 ppm of Si to evaluate the efficiency of the selected inhibitors at a dosage rate of 500 ppm. Under these conditions, GTMS-1, GTMS-2, and GTMS-4 exhibited dispersive effects, maintaining particles in suspension. In contrast, GTMS-3, GTMS-5, GTMS-6, and GTMS-8 resulted in visible aggregation, with particles settling at the bottom of the reactors. Notably, GTMS-7 showed a 100% recovery rate of Mg^{2+} in the solution, confirming its strong inhibitory performance under elevated scaling conditions. *(* note: The bottle test is limited to 150°C due to temperature and pressure rating for the reactors used)*

To further assess the suitability of the formulated inhibitors (Table 2) for controlling silica and metal silicate scaling in geothermal and HTHP applications, a tube-blocking method using DSL was employed. Each inhibitor was tested at one fixed concentration of 500 ppm and at elevated temperatures of 175°C and 200°C. The brine used was supersaturated with silica/Mg-silicate with a saturation index for Mg-silicate of approximately 6.47 at 175°C. The brine was passed through a 1 mm × 3 m coil at a flow rate of 2mL/

min under testing conditions of 175°C and ~500 psi to determine the blank scaling time [Table 1, Figure 5]. A slower brine flow rate of 2mL/min than the standard 10mL/min used in DSL tests was selected to increase residence time and promote silica/ silicate formation, which occurs more slowly than other common inorganic scales. Differential pressure was monitored over time to determine the tube blocking time without inhibitor or chemical treatment. In the blank test, the pressure threshold of <1 psi was reached in approximately 40 minutes, establishing the baseline reference. Using this reference, each GTMS product (GTMS-1 to GTMS-10) was tested for 2 hours to evaluate its ability to prevent aggregation and deposition within the capillary tubing. Among the tested inhibitors, all the GTMS formulations besides GTMS-4 and GTMS-5 successfully met the performance criteria, demonstrating effective scale control under the tested conditions (Figure 5).

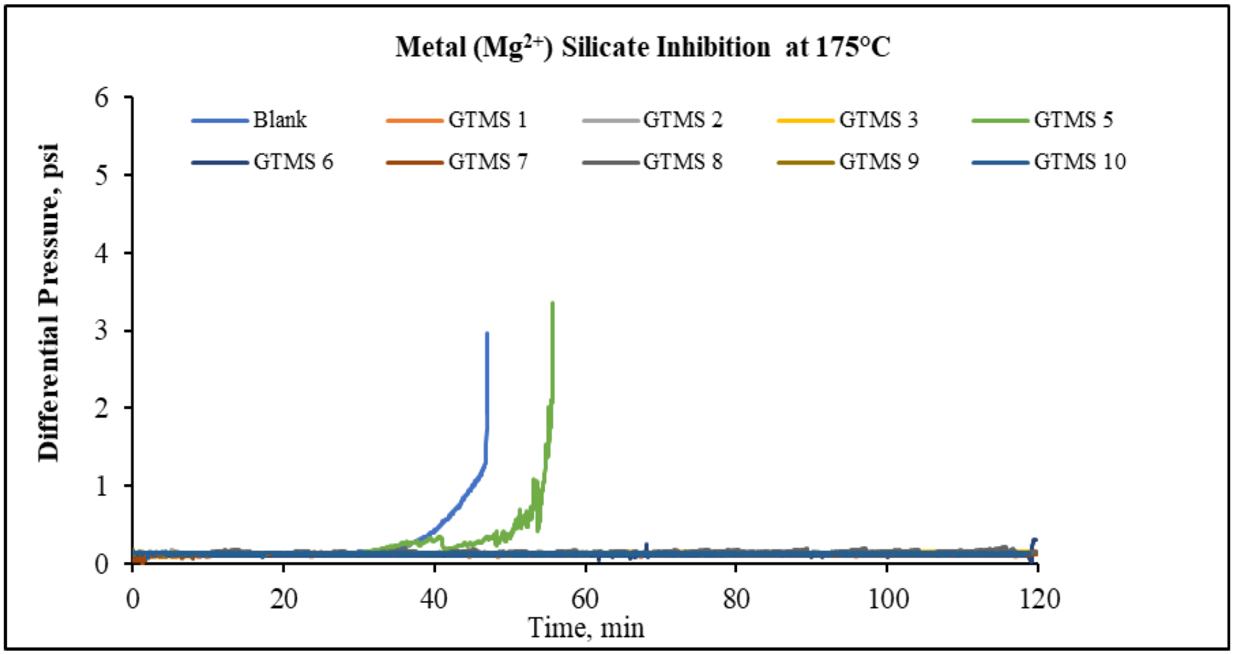

Figure 5—Evaluation of silica and magnesium silicate inhibition at 175°C with flow through system. Inhibitors' concentration is maintained at 500 ppm of formulated products

To evaluate the performance of the formulated inhibitors under more extreme conditions representative of high-enthalpy geothermal systems, tests were conducted at approximately 200°C using the same supersaturated brine (Mg-silicate saturation index ~7.18). At this temperature, the flow rate was increased to 10 mL/min, consistent with standard DSL testing protocols. During the blank run, frequent drops in differential pressure, from 1.8 psi to baseline 0.8 psi, were observed over the test time, indicating the formation of loosely bound precipitate aggregates that were intermittently dislodged from the surface of the reaction coil. A blank time of 15 minutes was established as the reference point, corresponding to the appearance of the first pressure peak (DP > 1 psi) (Figure 6). Each GTMS product (GTMS-1 to GTMS-10) was tested individually at 500 ppm for a duration of one hour to assess their ability to prevent tube blockage. Among these, GTMS-4, GTMS-5, GTMS-6, and GTMS-8 failed to protect the coil, resulting in complete or partial blockage within 10–20 minutes and exhibiting frequent pressure drops throughout the test period. This behavior is exemplified by the gray line representing GTMS-4 in Figure 6. In contrast, GTMS-1 to GTMS-3, GTMS-7, and GTMS-9 to GTMS-10 demonstrated excellent control over silica/Mg-silicate deposition, maintaining coil integrity for the full one-hour test duration indicating that these products show more inhibitory action than dispersants (Figure 6).

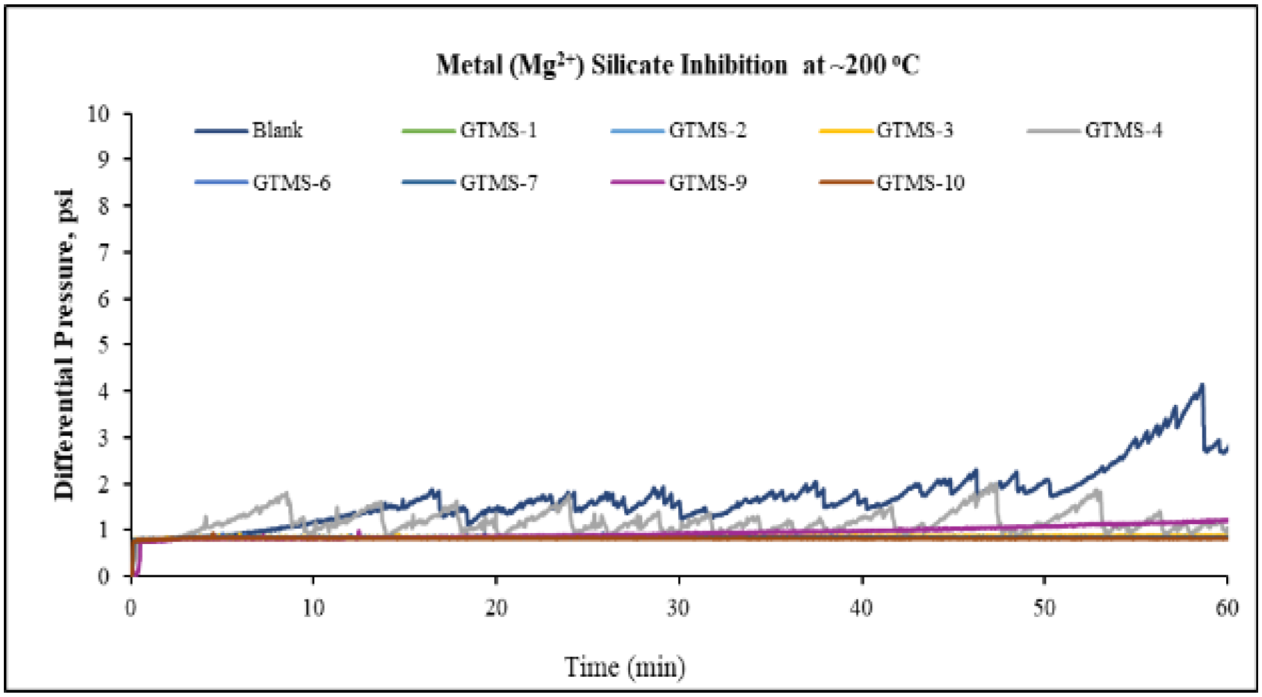

Figure 6—Screening of inhibitors' performance for silica and magnesium silicate control at ~200°C with flow through system. Inhibitors' concentration is maintained at 500 ppm of formulated products

From the initial screening with the flow-through system, at least six of the inhibitors were identified as suitable candidates for controlling silica and metal silicates at ultra-high temperatures (200°C). Among these, GTMS-2, GTMS-7, and GTMS-10, demonstrating strong performance in both static and dynamic tests, were selected for further evaluation. Minimum Inhibitory Concentrations (MICs) were determined to identify the lowest effective dosage capable of controlling the same amount of deposition observed during the DSL screening. The same blank reference was used for consistency. Starting concentrations of 200 ppm were applied to the dosed brine and subsequently reduced to 100 ppm and 50 ppm, with each dosage level tested over a one-hour protection period (Figure 7). The results showed that GTMS-2 and GTMS-7 were highly effective, successfully controlling and delaying the deposition of Mg-silicate precipitates (SSP-predicted mass: 3.87 mg/L) at concentrations as low as 100 ppm. However, both failed at 50 ppm. GTMS-2 also failed at 100 ppm, requiring a minimum of 200 ppm to maintain effective scale control (Figure 7).

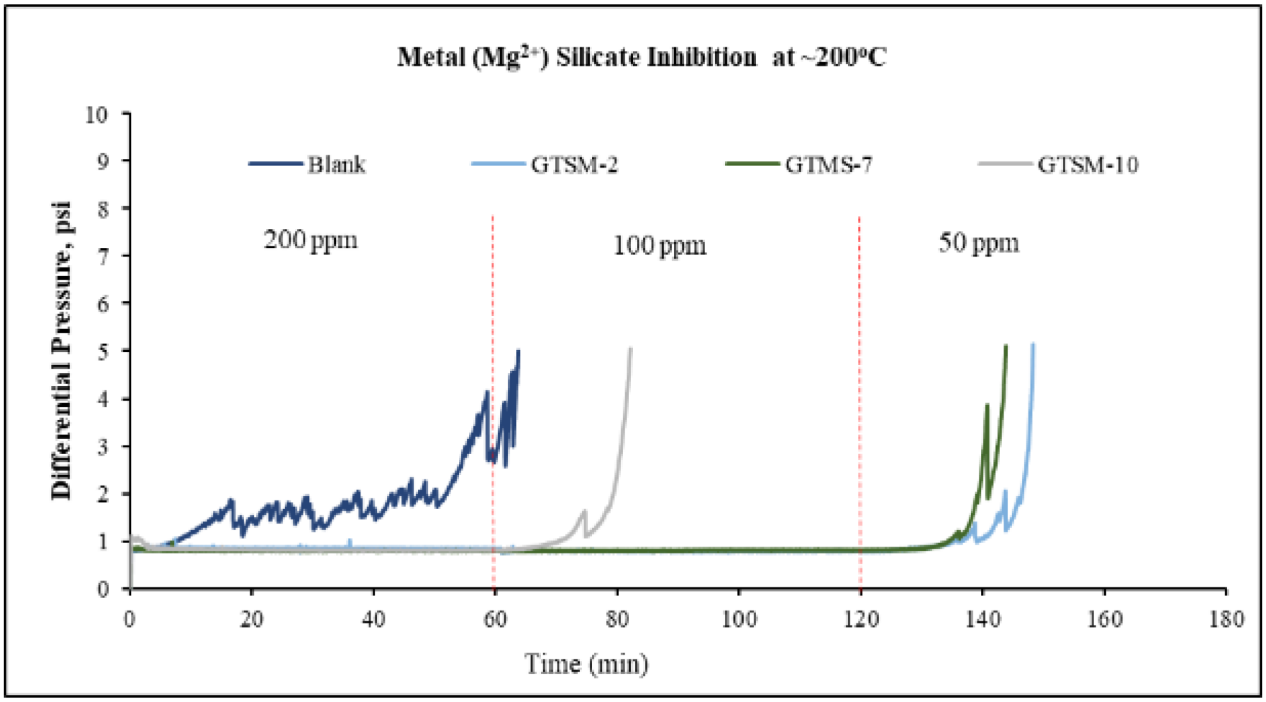

Figure 7—Identification of minimum effective dose for selected inhibitors to control silica and magnesium silicate at 200°C with flow through system

Inhibition of complex metal silicates with calcite

Calcium carbonate (calcite) is another type of scale that is commonly observed in oil and gas production systems and is a very dense and immensely adherent crystalline form of scale. The occurrence of this scale is very common in low (<90°C) and medium enthalpy (90-150°C) geothermal systems. Sometimes calcite is also observed as co-precipitates along with silica/metal silicates in high enthalpy (>150°C) geothermal systems (Andritsos, Karabelas and Koutsoukos, 1997). To evaluate scale control under these complex conditions, the same set of inhibitors (GTMS-1 to GTMS-11, each at 500 ppm) was tested using a batch method with brine supersaturated with silica, Mg-silicate, and calcite (saturation indices: Mg-silicate = 14.54, calcite = 1.63). Among the tested products, GTMS-1, GTMS-2, GTMS-3, GTMS-4, and GTMS-11 demonstrated effective control of mixed scale types through either inhibitory or dispersive mechanisms. For example, GTMS-4 and GTMS-11 promoted dispersion of precipitates, while GTMS-1 and GTMS-2 maintained clear solutions without visible precipitation for up to four hours (Figure 8b). After 24 hours, all treated samples exhibited dispersed particles and some deposition at the bottom of the reactors, indicating a decline in inhibitor efficiency over time (Figure 8c). Upon cooling the samples to approximately 100°C, the solid precipitates redissolved into solution (Figure 8d), suggesting the formation of metastable scale phases following chemical interaction. This behavior supports the application of rapid cooling techniques—where temperature is instantly reduced by ~100°C—to mitigate scale deposition in geothermal systems. Such rapid cooling methods have been implemented in geothermal fields including Svartsengi, Namafjall, and Nesjavellir in Iceland, and Olkaria in Kenya, where temperature drops from ~160°C to ~70°C have proven effective in controlling amorphous silica deposition (Arnórsson, 2000). The use of these newly developed inhibitors in conjunction with rapid cooling may offer a promising strategy for reducing complex inorganic scale formation in geothermal operations.

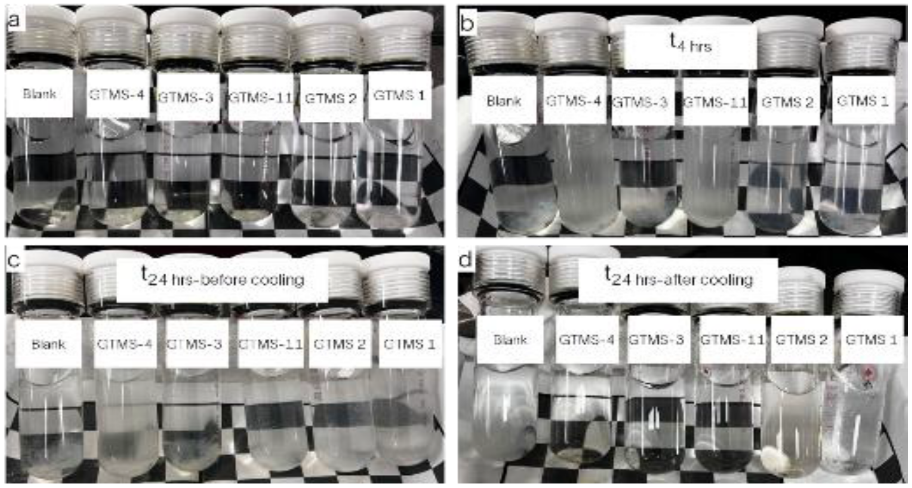

Figure 8—Evaluation of inhibition of complex scale with silica, magnesium silicate and calcite at 150°C with static method for 24 hours: a. right after mixing anion and cation b. visual observation at t_{4hrs} c. visual observation at 24 hours before cooling d. visual observation at 24 hours after cooling down to the room temperature.

Following the evaluation of inhibitors for silica and metal silicate control, selected formulations were further tested under complex scaling conditions to assess their effectiveness against mixed scales. GTMS-2 and GTMS-7—identified as top performers for metal silicate inhibition—were selected for testing with brine supersaturated with both metal silicate and calcite. The tube-blocking method using the DSL system was employed to determine the MICs required to prevent deposition. A blank reference time of approximately 6 minutes was established using the same brine as in the batch tests. When GTMS-2 was dosed at 500 ppm, the differential pressure (ΔDP) remained close to baseline throughout the one-hour test, indicating complete inhibition of scale formation. However, when the concentration was reduced to 400 ppm, a slight increase in ΔDP (~1 psi) was observed toward the end of the test, suggesting this dosage is near the MIC threshold for GTMS-2 (Figure 9). Similarly, GTMS-7 maintained baseline ΔDP at a concentration of 300 ppm, effectively preventing deposition. When the dosage was further reduced to 200 ppm, the ΔDP exceeded the 1 psi threshold within approximately 12 minutes, indicating that the MIC for GTMS-7 under these conditions is around 300 ppm.

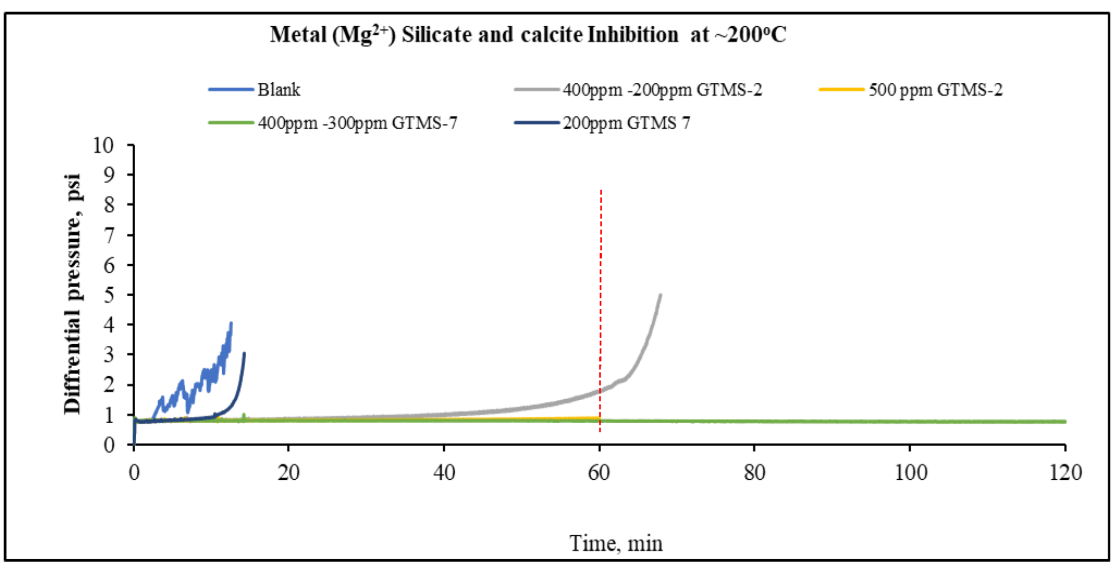

Figure 9—Identification of minimum effective dose for selected inhibitors to control silica, magnesium silicate and calcite at 195°C with flow through system

Proposed flow-through method for silica silicate evaluation up to 250°C

To overcome the limitations of conventional scale testing methods, we have recently designed and developed a novel high-pressure, HTHP flow-through apparatus capable of operating at temperatures up to 250 °C. This system enables the evaluation of scale formation and inhibition under realistic geothermal conditions (Figure 10). Compared to industry-standard methods such as DSL, this new apparatus offers significantly higher residence volumes (~25 ml) and extended residency time. In one configuration, a flow rate of ~0.01 mL/min was maintained to achieve a constant pressure of 850 psi at 250 °C, resulting in a maximum residence time of approximately 40 hours. This extended residence time is particularly advantageous for capturing the slow kinetics associated with silica and silicate scale formation. Method validation and product evaluation using this apparatus are currently underway, as shown in Figure 10. The system represents a significant advancement in scale testing technology, though the development of inhibitors capable of maintaining efficacy at temperatures beyond 250 °C remains a challenge and will require further innovation in chemical design and qualification.

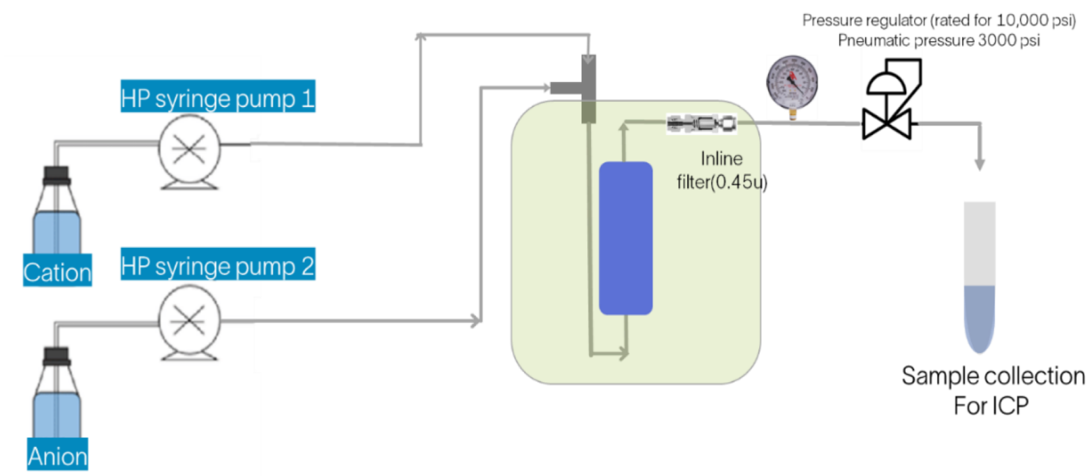

Figure 10—Proposed HTHP flow loop apparatus for silica/ silicate scales

CONCLUSIONS AND RECOMMENDATIONS

In this study, a comprehensive evaluation of selected products, including polymers, phosphonates, organic acids, and polyphosphonates, was conducted using both batch and tube-blocking methods across a range of temperatures, brine compositions, saturation indices, and dosage levels. At least three products—GTMS-2, GTMS-7, and GTMS-10—were found to be effective in controlling silica and metal silicate scaling at temperatures up to ~200°C using nominal dosage rates. Furthermore, GTMS-2 and GTMS-7 demonstrated strong performance against complex scaling conditions involving coexisting metal silicate and calcite. However, the required dosage rates were comparatively higher due to increased scaling severity and elevated saturation ratios resulting from changes in brine composition and ion concentrations.

Although the selected products were qualified using industry-standard methods and equipment, these approaches may not fully capture the kinetics of slow-forming amorphous silica and silicate scales. Additionally, the temperature limitations of current testing apparatus restrict the ability to simulate scale formation and inhibition mechanisms under the extreme conditions found in geothermal power plants, where temperatures can exceed 250°C. To address these limitations, flow-through HTHP apparatus that is designed to operate at temperatures up to 250°C has been developed. This may enable more realistic evaluations of scale inhibition performance under geothermal conditions. The details of this new methodology and apparatus will be presented in future research publications and technical conferences. Even though we have a method to analyze inhibitors at such temperature, the development of inhibitors to sustain and maintaining efficacy at temperatures beyond 200°C remains a significant challenge. This will require continued innovation in product design, formulation, and qualification processes to meet the demands of ultra-high-temperature geothermal applications.

Nomenclature

ΔP	Differential pressure
ASP	Alkaline-Surfactant-Polymer
DSL	Dynamic scale loop
GTMS	Geothermal metal silicate inhibitor
HTHP	High-temperature high-pressure
ICP-OES	nductively coupled plasma optical emission spectrometry
SR	Saturation ratio
SI	Saturation Index
SSP	ScaleSoftPitzer
WAG-EOR	Water-Alternating-Gas enhanced well recovery

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